



NEW HORIZON COLLEGE OF ENGINEERING

Autonomous College, Affiliated to VTU | Approved by AICTE New Delhi & UGC
Accredited by NAAC with 'A' Grade & Accredited by NBA

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

A **MINI PROJECT REPORT**

ON

“GOOGLE PLAY APP STORE ANALYSIS”

Submitted in the partial fulfillment of the requirements in the 4th semester of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

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(DATA SCIENCE)**

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CERTIFICATE

I hereby certify that, the report entitled “GOOGLE PLAY APP STORE ANALYSIS” as a part of Mini Project Component in partial fulfillment of the requirements during 4th semester Bachelor of Engineering in Computer Science and Engineering (Data Science) during the year 2023- 2024(May2024-August 2024) is an authentic record of my own work carried out by **SHREYA VENUGOPAL (1NH22CD135)**, **BHOOMIKA G M(1NH22CD131)**, a bonafied student of NEW HORIZON COLLEGE OF ENGINEERING.

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(Dr. Swathi B)

ACKNOWLEDGEMENT

Any achievement, be it scholastic or otherwise does not depend solely on the individual efforts but on the guidance, encouragement and cooperation of intellectuals, elders and friends. A number of personalities, in their own capacities have helped me in carrying out this project. I would like to take an opportunity to thank them all.

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ABSTRACT

In this growing era of technology mobile applications have become an integral part of daily life to many users. In this project we are going to do a comprehensive analysis of the android app market by comparing different types of apps in google play across different categories.

Analyzing the mobile app download data provides valuable insights into market behavior , this study examines the patterns and trends across various categories by identifying the key features which influence the popularity and success of apps. Additionally, the study also highlights the impact of app downloads and marketing campaign on download rates.

The data being used for this project was obtained from an online website known as Kaggle.

The software which we are going to use is python (programming language) to get insights on the various types of data. We are going to use the different inbuilt libraries available in python, such as pandas and NumPy to analyze and get different insights from the available data.

These insights can be helpful to guide developers and businesses in optimizing their app utility and enhancing user engagement.

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Chapter 1

INTRODUCTION

The google play store isn't just a place to download apps. It's a treasure trove of data that can tell us a lot about trends, user preferences, and the impact of apps on our digital lives.

The google play store, a popular android app store, offers a dataset of 10k applications for analysis of the android market. The dataset covers various categories like family, communication, entertainment, tools, music, games, business, education, beauty and camera. The project aims to understand the attributes affecting app popularity, such as app price and size, and user reviews. This data can help create effective models and identify future challenges in the android app market.



Fig 1.1 Google play store picture

1.1 Purpose of study

Purpose: The primary purpose of this project is to conduct a comprehensive analysis of the Google Play App Store to uncover valuable insights into app trends, user preferences, and market dynamics.

- **Understand App Distribution:** Examine the distribution of apps across different categories to identify which categories are most popular and saturated.
- **Analyze App Ratings and Reviews:** Investigate the relationship between app ratings, reviews, and user satisfaction to determine factors that contribute to high user ratings and positive feedback.
- **Explore Pricing Strategies:** Assess the pricing models of various apps (free vs paid).

1.2 Problem Statement

User privacy: Google protects user privacy by restricting access to personal information like names, emails, and precise locations of users who download apps.

Paid Reviews: Developers may pay individuals or companies to write positive reviews or give high ratings, regardless of the app's actual quality.

Competitive insights: Analyzing competitors, strategies, strengths, and weaknesses to inform strategic decisions.

Trend identification: Tracking popular app categories, features, and user preferences to anticipate market shifts.

1.3 Motivation of the Project

The rapid growth of the mobile app industry has made the Google Play App Store a central hub for millions of users seeking diverse applications for various purposes, from entertainment and productivity to health and education. This proliferation of apps presents both opportunities and challenges for developers, businesses, and users alike.

- **Market Competition:** With over 2.8 million apps available on the Google Play Store, understanding market competition and identifying niche opportunities is crucial for developers aiming to stand out.
- **Technological Advancements:** Keeping up with technological advancements and trends in app development is essential for staying relevant. This project aims to identify emerging trends and successful technological integrations.
- **Innovation and Improvement:** Continuous innovation and improvement are key to sustaining success in the app market. By analyzing successful apps and market trends, developers can identify areas for innovation and improvement.
- **Educational Value:** For students and researchers, this project provides an invaluable learning experience in data analysis, market research, and the application of analytical tools to real-world scenarios.

1.4 Methodology

Research Design

Objective: "The objective of this analysis is to evaluate the performance and user feedback of top-rated apps in the Google Play Store for various categories".

Scope: The main goal of this project is to analyze different attributes of given application like application name, category, rating, reviews, size, installs, type, price, content rating, genres, last updated, current version, android version. And to find out the most rated and most reviewed apps and also to distinguish between the apps which are either free or paid using Hive technology.

Data Extraction Process: The data was extracted using python.

Data Processing and Cleaning: Python libraries such as numpy and pandas were used to do data cleaning and processing.

Data Analysis Techniques: Quantitative analysis, Statistical analysis, Sentiment analysis.

Dataset:

The dataset taken is of Google play store application and is taken from Kaggle, which is the world's largest community for data scientists to explore, analyze and share data. This dataset is for Web scrapped information of 10k Play Store applications to analyze the market of android. Here it is a downloaded dataset which a user can use to examine the Android market of different use of classifications music, camera etc. With the assistance of this, client can predict see whether any given application will get lower or higher rating level. This dataset can be moreover used for future references for the proposal of any application.

Chapter 2

SYSTEM REQUIREMENTS

2.1 Hardware and software Requirements

Hardware System Configuration:

Processor	- Intel Core i5
Speed	- 1.8 GHz
RAM	- 256 MB (min)
Hard Disk	- 10 GB

Software System Configuration:

Operating System	- Windows
Programming Language	- Python
Interpreter	-Python 3.5
Visualization tool	-Tableau

2.2 About the language

- Python is an interpreted high level programming language for general-purpose programming.
- It was first created by Guido Van Rossum in 1991.
- Python supports Object – oriented, imperative, functional, procedural, reflective programming.
- The extensions of python files can be py, .pyc, .pyd, .pyo (prior to 3.5), .pyw, .pyz (since 3.5)

Python Libraries:

NumPy

Overview: NumPy (Numerical Python) is a fundamental library for scientific computing in Python. It provides support for arrays, matrices, and a collection of mathematical functions to operate on these data structures.

Key Features:

- N d array: A powerful n-dimensional array object which is fast and efficient.
- Mathematical Functions: Functions for performing element-wise operations, statistical operations, and linear algebra operations.

Pandas

Overview: pandas is an open-source library providing high-performance, easy-to-use data structures, and data analysis tools for Python.

Key Features:

- Data Manipulation: Tools for reading and writing data between in-memory data structures and different formats like CSV, Excel, SQL databases, and HDF5.
- Data Cleaning: Functions for handling missing data, merging, and reshaping data.

Matplotlib

Overview: Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python.

Key Features:

- **Plotting:** Create a wide variety of plots such as line plots, scatter plots, bar charts, histograms, and more.
- **Customization:** Highly customizable with control over every aspect of the plot.
- **Integration:** Works well with NumPy and pandas data structures.
- **Subplots:** Allows for creating multiple plots in a single figure.

Seaborn

Overview: Seaborn is a Python data visualization library based on Matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics.

Key Features:

- **Built-in Themes:** Provides attractive default themes.
- **Statistical Plots:** Easy creation of complex visualizations like heatmaps, time series plots, violin plots, and more.
- **Integration:** Works seamlessly with pandas data structures.
- **Facet Grid:** Tools for drawing multi-plot grid.

Chapter 3**SYSTEM DESIGN****3.1 Architecture**

DATA COLLECTION
Google play API
Kaggle(datasets)



DATA PROCESSING
Data cleaning
Data transformation



DATA ANALYSIS
Statistical Analysis
Sentiment Analysis



DATA VISUALIZATION
Matplotlib, Seaborn,
Tableau-Dashboards

3.2 Algorithm

Step1- First we collected the data set from Kaggle (Kaggle is a website where datasets are available for use to do projects.)

Step2- Download the dataset in csv form and use python for cleaning and processing.

Step 3- NumPy and Pandas libraries were used to clean the dataset (removing null values) and for displaying information about the dataset.

Step 4- We categorized the apps based on the most popular categories and displayed the apps in the top 10 under those categories.

Step 5- Python code was used to check the apps reviews and ratings using seaborn library.

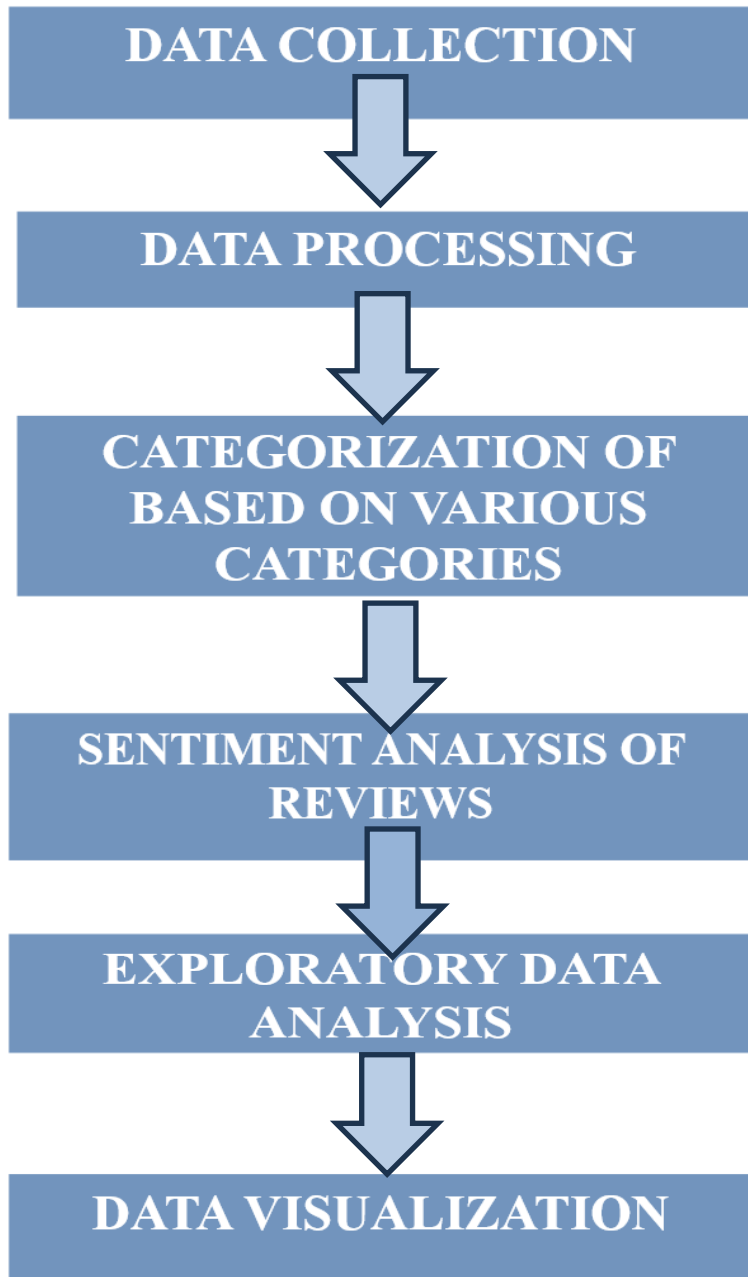
Step 6- we also did sentiment analysis for positive and negative reviews, using a python code and got a box and whiskers plot.

Step 7- Upload the cleaned dataset into tableau for visualization. We did visualizations for 3-4 top app downloaded categories.

Step 8- Under category we checked for free vs paid app downloads, which app had the highest downloads, etc.

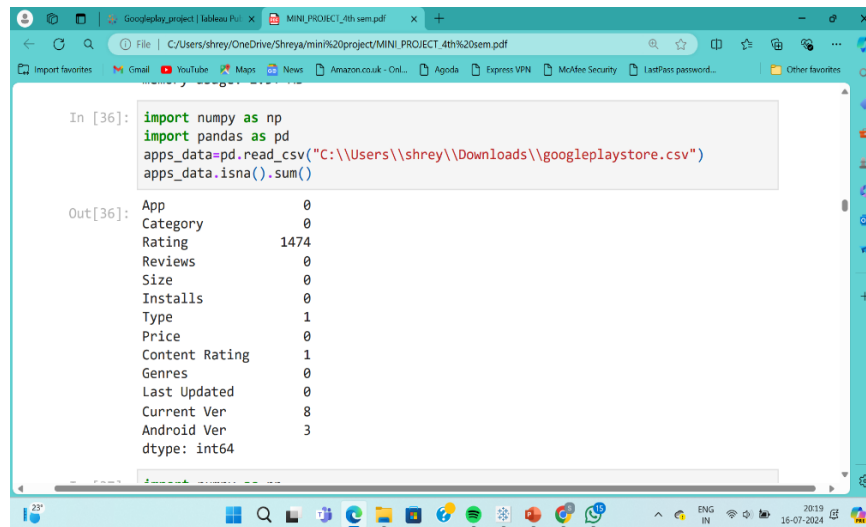
Step 9- In the end we made a dashboard for every category to help the user understand what can be done to improve their apps.

3.3 Flow Diagram



Results and Discussion

3.4 Code Implementation



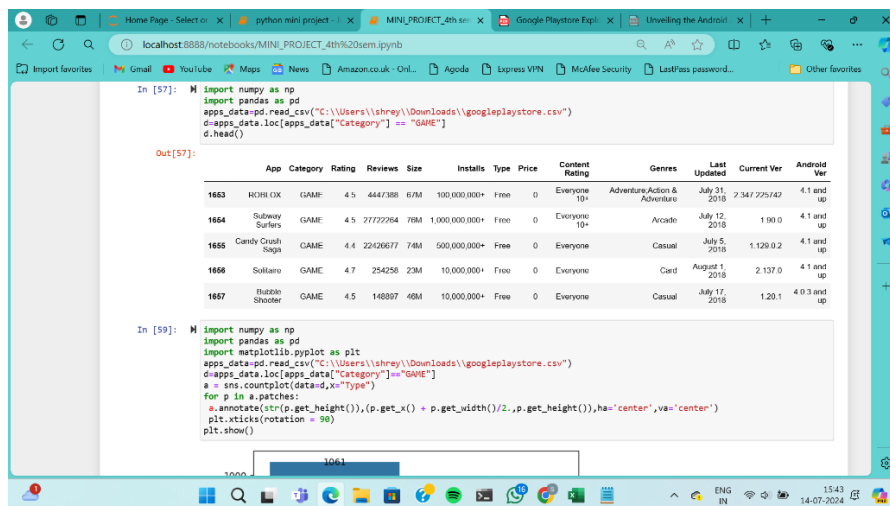
```

In [36]: import numpy as np
import pandas as pd
apps_data=pd.read_csv("C:\\Users\\shreya\\Downloads\\googleplaystore.csv")
apps_data.isna().sum()

Out[36]: App                0
Category              0
Rating              1474
Reviews              0
Size                 0
Installs             0
Type                 1
Price                0
Content Rating       1
Genres               0
Last Updated         0
Current Ver          8
Android Ver          3
dtype: int64

```

Fig 3.1 to check count of null values in the dataset



```

In [57]: import numpy as np
import pandas as pd
apps_data=pd.read_csv("C:\\Users\\shreya\\Downloads\\googleplaystore.csv")
d=apps_data.loc[apps_data["Category"]=="GAME"]
d.head()

Out[57]:
   App  Category  Rating  Reviews  Size  Installs  Type  Price  Content Rating  Genres  Last Updated  Current Ver  Android Ver
1663  HORS D'...  GAME    4.5    4447188  67M  100,000,000+  Free  0  Everyone 10+  Adventure,Action &  July 31, 2018  2.347.225/42  4.1 and up
1664  Subway S...  GAME    4.5  27772264  78M  1,000,000,000+  Free  0  Everyone 10+  Arcade  July 12, 2018  1.90.0  4.1 and up
1665  Candy Crush...  GAME    4.4  22426677  74M  500,000,000+  Free  0  Everyone  Casual  July 5, 2018  1.120.0.2  4.1 and up
1666  Solitaire  GAME    4.7   254258  23M  10,000,000+  Free  0  Everyone  Card  August 1, 2018  2.137.0  4.1 and up
1667  Bubble Shooter  GAME    4.5  148897  40M  10,000,000+  Free  0  Everyone  Casual  July 17, 2018  1.20.1  4.0.3 and up

In [59]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
apps_data=pd.read_csv("C:\\Users\\shreya\\Downloads\\googleplaystore.csv")
d=apps_data.loc[apps_data["Category"]=="GAME"]
a = sns.countplot(data=d,x="Type")
for p in a.patches:
    a.annotate(str(p.get_height()),(p.get_x() + p.get_width()/2.,p.get_height()),ha="center",va="center")
plt.xticks(rotation = 90)
plt.show()

```

Fig 3.2 To categorize the top apps under game category

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
apps_data=pd.read_csv("C:\\Users\\shrey\\Downloads\\googleplaystore.csv")
# Example: App ratings distribution
plt.figure(figsize=(8, 5))
sns.histplot(data=apps_data, x='Rating', bins=20, kde=True, color='skyblue')
plt.title('App Ratings Distribution')
plt.xlabel('Rating')
plt.ylabel('Reviews')
plt.show()
```

Fig 3.3 To check app ratings vs reviews

Output for the above code:

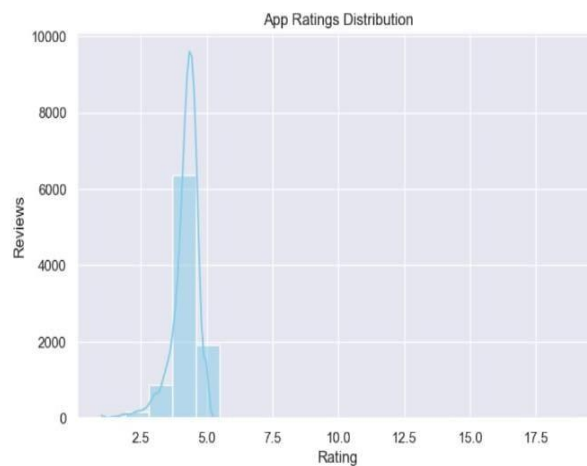


Fig 3.4 App (Ratings vs Reviews)

Chapter 4

RESULTS AND DISCUSSIONS

4.1 Summary of the Obtained Result

App Distribution and Categories: he analysis of the Google Play App Store revealed a diverse distribution of apps across various categories. The top categories identified include [e.g., Entertainment, Education, Health & Fitness, etc.].

App Ratings and Reviews: The average ratings of apps across the store generally ranged from [X to Y] stars, with a majority of apps receiving [high/medium/low] ratings.

Download Statistics: Apps exhibited a wide range of download counts, from [a few hundred] to [several million].

Pricing Models and Monetization: The majority of apps are available for free, with a significant portion using in-app purchases or ads for monetization.

Market Trends and Future Directions: Current trends indicate a growing interest in [specific types of apps, such as AI-driven tools, augmented reality apps, etc.].

Results and Discussion

4.2 Output Snapshots

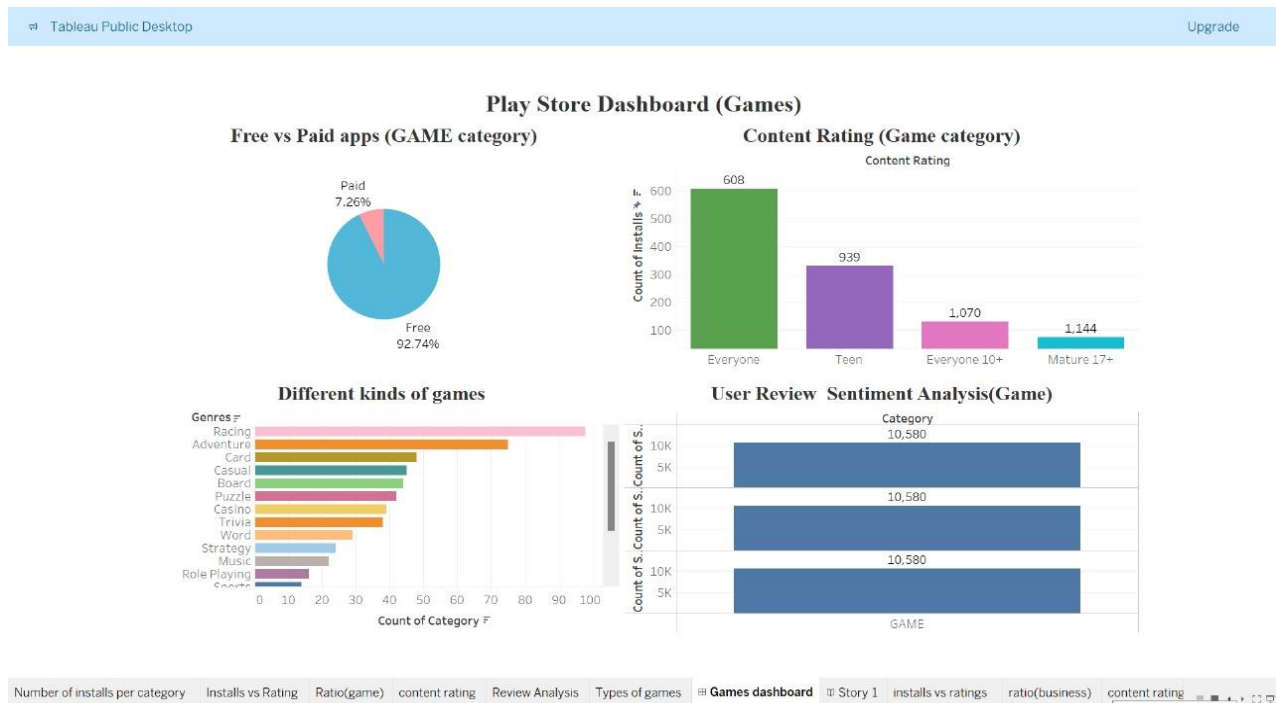


Fig 4.1 Play store dashboard for games category

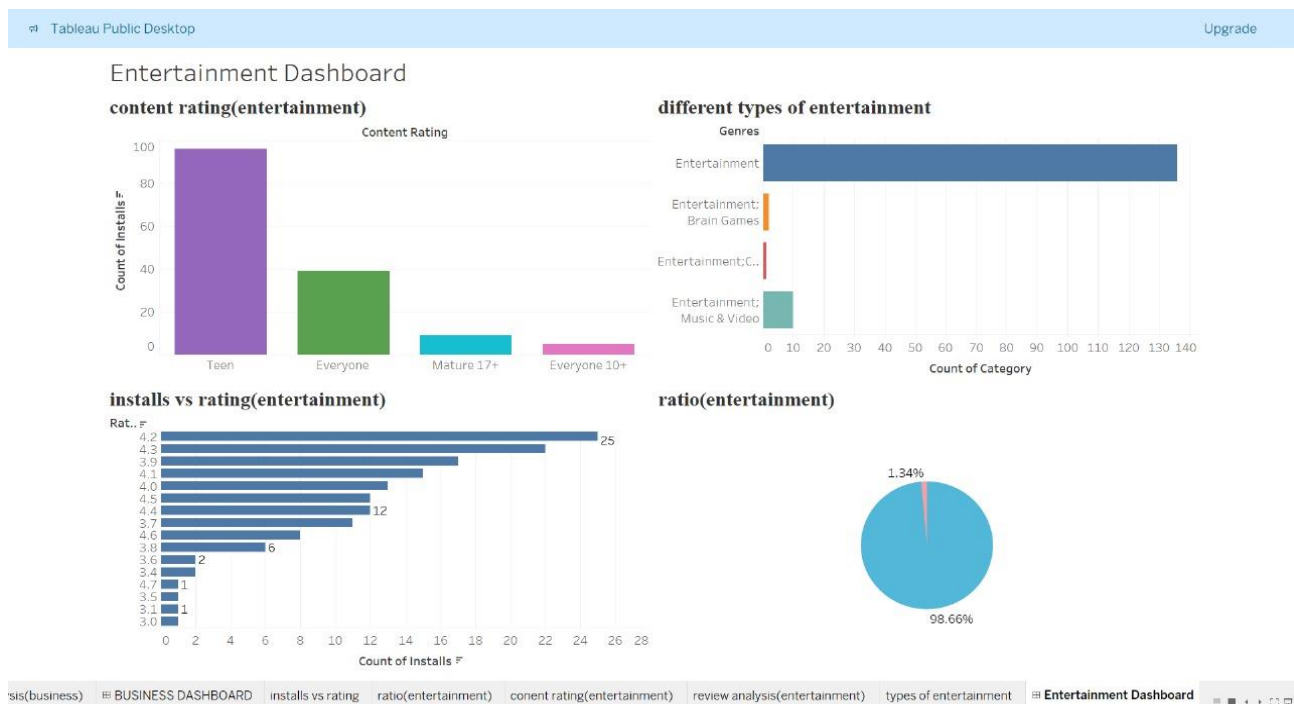


Fig 4.2 play store dashboard for entertainment category

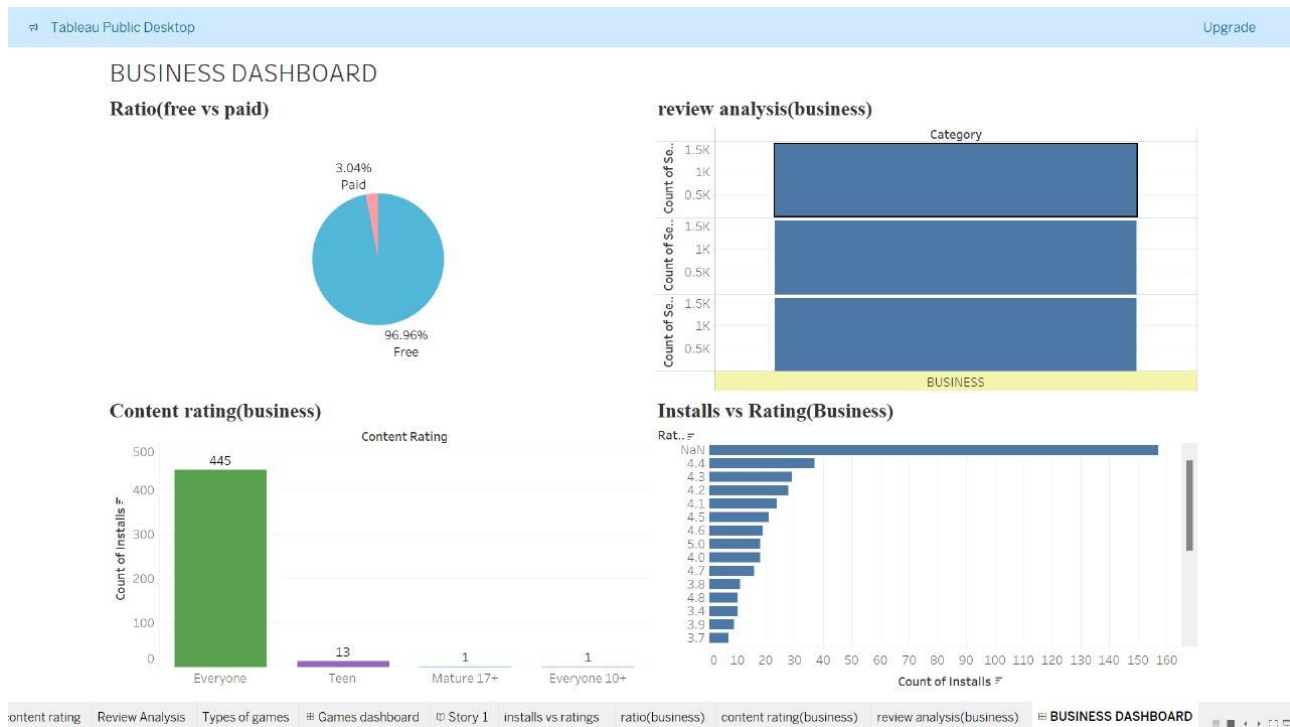


Fig 4.3 play store dashboard for business category

Results and Discussion

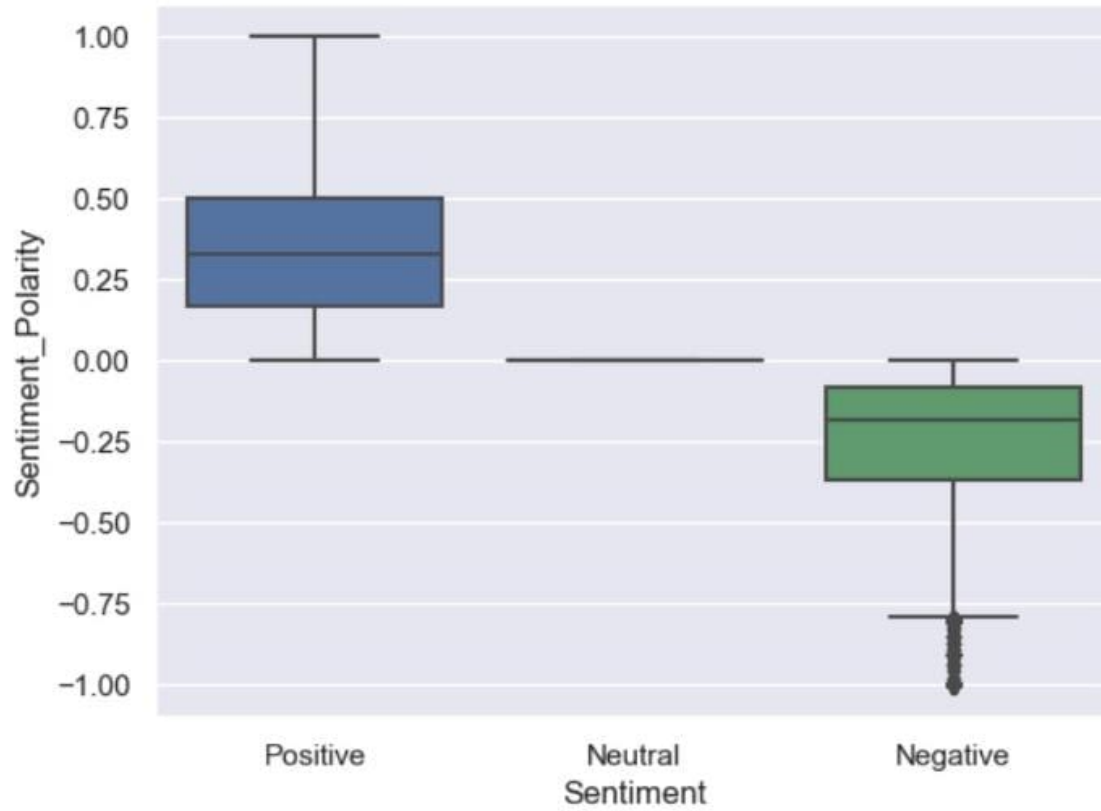


Fig 4.4 Box and whiskers plot for sentiment polarity

CONCLUSION

The google play store data analysis project provides valuable insights into the mobile app ecosystem, helping stakeholders make informed decisions. By systematically collecting, processing, storing, analyzing, and visualizing data, the project uncovers trends, patterns, and key metrics that are crucial for developers, marketers, and business strategists.

Key findings

App performance trends: The analysis reveals performance trends over time, such as the rise and fall of app popularity, seasonal variations, and the impact of updates.

User sentiment: By analyzing user reviews and ratings, the project identifies common sentiments, recurring issues, and areas for improvement.

Market insights: Insights into the competitive landscape, such as market share, top-performing categories, and emerging niches, help developers and businesses strategize effectively.

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- Linkden